Vector Spaces IV: Quotient Spaces and Counting over Finite Fields: Answer Key

Problem sheet for the Day 5 class · CSIR NET previous year questions and exam-pattern problems

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(A)	(D)	(C),(D)	10	9	24	13	35	4	(A),(B),(C)

Hints and short solutions

1. **(A)** CSIR NET Dec 2016 $(8-1)(8-2)(8-4) = 7 \cdot 6 \cdot 4 = 168$: choose independent columns one at a time.
2. **(D)** CSIR NET Dec 2015 $\det = 59(91 \cdot 31 - 31 \cdot 29) = 59 \cdot 31 \cdot 62 = 2 \cdot 31^2 \cdot 59$. The matrix is invertible mod p for every prime except $2, 31, 59$.
3. **(C),(D)** CSIR NET Dec 2014 $\det A = -416 = -2^5 \cdot 13$. An integer inverse needs $\det = \pm 1$. (So A is invertible over $\mathbb{F}_p$ exactly for $p \neq 2, 13$.)
4. **10** Exam pattern $25 - 15 = 10$ (the quotient is isomorphic to the skew-symmetric matrices).
5. **9** Exam pattern $\dim(V/W) = 2$, so it has 3^2 elements (the cosets).
6. **24** Exam pattern $|GL_2(\mathbb{F}_3)| = (9-1)(9-3) = 48$, and $\det$ maps onto $\mathbb{F}_3^*$ with equal fibres: $48/2 = 24$.
7. **13** Exam pattern $(3^3 - 1)/(3 - 1) = 13$: each line has 2 non-zero vectors.
8. **35** Exam pattern $\frac{(2^4-1)(2^4-2)}{(2^2-1)(2^2-2)} = \frac{15 \cdot 14}{3 \cdot 2} = 35$.
9. **4** Exam pattern Need $\operatorname{tr} A = 0$, $\det A = 0$: $d = a$, $bc = a$. $a = 0$: 3 choices of (b, c); $a = 1$: 1. Total 4.
10. **(A),(B),(C)** Exam pattern $p \mapsto p(1)$ is onto $\mathbb{R}$ with kernel W; so $\dim W = 3$.